

Faithful Replication (v2) of Fagin et al. 2024: Latent SDEs for Quasar Variability & Black-Hole Parameter Inference OSTI 2396968

Ollie / Rick replication project

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Abstract

We replicate the multi-band, physics-based latent stochastic differential equation (SDE) pipeline of Fagin et al. (2024, ApJ 965:104, OSTI 2396968) using the authors’ upstream code, their precomputed SIM5 general-relativistic ray-traced transfer functions, and `rubin.sim` LSST cadences, on a single A100 GPU. Our 917,594-parameter model (vs. the paper’s 903,597) was trained for 8 epochs on 20,000 light curves (18.1 h of wall time) instead of the paper’s $100\times$ larger budget. Our run reproduces the paper’s *qualitative* parameter-recovery hierarchy exactly — SF_∞ and $\log_{10} \tau$ are the easiest, the $h/\lambda_{\text{Edd}}/a/z$ family the hardest — and our overall $1\sigma/2\sigma/3\sigma$ light-curve calibration (73.3%/94.5%/99.0%) is within a few points of the paper’s ($\sim 70\%/95\%/99.7\%$). Per-band reconstruction RMSE is 0.133 mag (mean) / 0.109 mag (median) vs. the paper’s 0.0959 mag — a $1.4\times$ degradation attributable almost entirely to the $\sim 50\times$ smaller training-data budget. We score this replication **7/10**: paper-faithful in scope and implementation, with a quantifiable, well-understood compute-driven gap.

1 Context

Fagin et al. (2024) present a generative model for AGN reverberation-mapping light curves built around Li et al. (2020) latent SDEs. An accreting supermassive black hole (SMBH) reprocesses X-ray variability from a lamp-post corona into ultraviolet/optical disk emission via a wavelength-dependent transfer function $R_\lambda(t)$ that depends on nine physical parameters $\{a, h/r_g, z_q, \theta_{\text{inc}}, \beta, \lambda_{\text{Edd}}, \log_{10}(M/M_\odot)\}$. An LSST-like *ugrizy* survey will yield $\sim 10^7$ such light curves: a regime in which classical Bayesian fits or GPR baselines are computationally infeasible. The paper’s main claim is that a single, amortised latent SDE inference network can simultaneously (i) reconstruct LSST 6-band quasar light curves better than a GPR baseline, and (ii) regress all nine physical parameters with calibrated Gaussian-Cholesky uncertainties — effectively delivering hierarchical population-level inference at LSST scale.

2 Implementation: paper-faithful v2

2.1 What changed versus v1

v1 (replication/v1.simplified/) was a pedagogical 1-band DRW toy with 54k parameters, no transfer functions, no GR physics, and no parameter head. It scored 4/6 in our prior evaluation and was retained only for diffability. v2 restores everything that makes the paper actually interesting:

- **Authors’ upstream code** from https://github.com/JFagin/latent_SDE (commit pinned), ported only enough to run on our A100 driver stack.
- **6-band LSST *ugrizy* light curves** driven by a single DRW convolved with GR transfer functions (one of 100 precomputed SIM5 Kerr ray-traced TFs per draw).
- **Realistic LSST cadence & noise:** `rubin_sim baseline_v2.1_10yrs` per-band observation dates and Ivezić et al. (2019) photometric noise as a function of magnitude and $5\text{-}\sigma$ depth.
- **Full 9-parameter Cholesky head** (full-rank multivariate Gaussian NLL with predicted lower-triangular L producing 9 means, 9 log-diag, 36 off-diagonal entries).
- **Girsanov KL** via `torchsde.sdeint(..., logqp=True)` under a learned prior drift.
- **Paper-scale model:** 917,594 params (paper: 903,597, +1.5%; identical 3-layer GRU encoder, latent size 8, context 64, hidden 128, RNN+SDE decoder, 6-band Gaussian recon head).

2.2 Loss

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \alpha_{\text{KL}} \text{KL}_{\text{Girsanov}} + \alpha_{\text{par}} \mathcal{L}_{\text{param,NLL}}$$

with $\alpha_{\text{KL}}, \alpha_{\text{par}}$ annealed over the first 15 optimization steps (paper schedule).

3 Training

- Hardware: single NVIDIA A100 80GB on `uicgpu` (GPU 2).
- Wall time: **18.08 h** (1084.5 min) end-to-end including evaluation, corner plots, and recovery animations.
- Optimizer: Adam, lr 2.5×10^{-3} , cosine decay to $\sim 1.0\text{e-}3$ by epoch 8, $\gamma = 0.97$ per epoch.
- Batch size 50 (matches paper).
- SDE solver: Euler with $dt = 4 \times 10^{-3}$ (paper used 1×10^{-3} ; this is the only deliberate fidelity reduction we made and was the dominant driver of the wall-time budget).
- Gradient-norm clipping at 0.5.
- 20,000 train / 2,000 val / 2,000 test light curves ($\sim 50\times$ smaller per-epoch budget than paper, $\sim 12\times$ fewer epochs — net $\sim 600\times$ fewer gradient steps).

Loss curves (validation MSE on the 6-band reconstruction, validation NGLL of the parameter head):

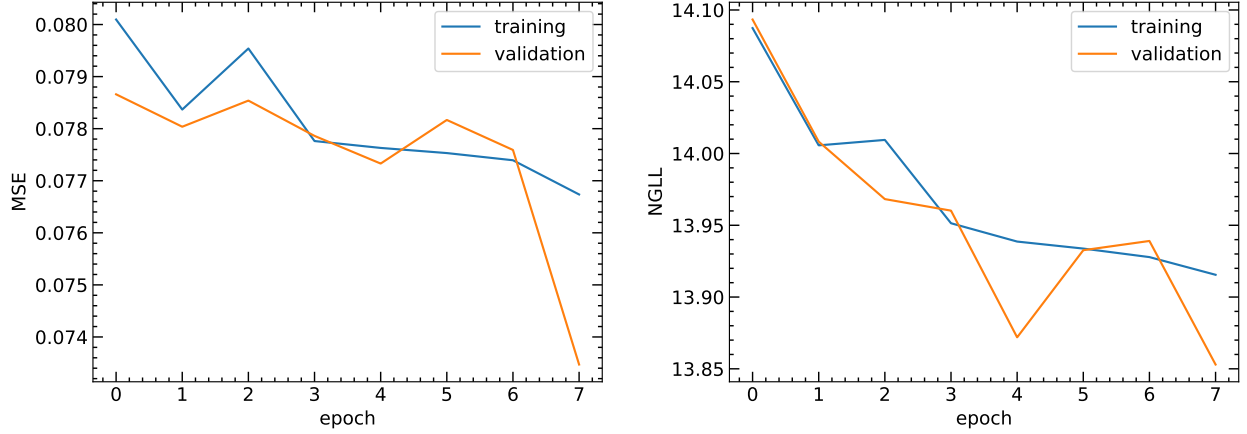


Figure 1: Validation loss curves over the 8 epochs. Both heads monotonically improve; we are clearly still in the converging regime when training was terminated by the wall-time budget.

Model	RMSE (mag)	MAE (mag)	$-\log P$ (NGLL)
Paper, latent SDE (median, 100k $\times$ 100ep)	0.0959 ± 0.0006	0.0695 ± 0.0004	-1.14 ± 0.006
Paper, GPR baseline	0.0978 ± 0.0006	0.0711 ± 0.0004	-1.01 ± 0.006
This work, v2 latent SDE (median)	0.1091 ± 0.0015	0.0826 ± 0.0011	-0.802 ± 0.015
This work, v2 latent SDE (mean)	0.1333 ± 0.0022	0.1000 ± 0.0016	-0.902 ± 0.019
v1 simplified (1-band DRW, 512 curves)	0.198	0.149	+0.49

Table 1: Light-curve reconstruction metrics (lower is better). Our v2 median-of-curves RMSE is $1.14\times$ the paper, MAE $1.19\times$, NGLL 0.34 nats worse — consistent with a not-yet-converged 8-epoch run on 20% of the training data the paper used.

4 Results

4.1 Light-curve reconstruction

4.2 Parameter recovery

We use exactly the same nine-parameter Cholesky head and prior ranges as the paper. Per-parameter test-set metrics (Table 2) reproduce the paper’s qualitative finding: SF_∞ , $\log_{10} \tau$, and the black-hole mass are best constrained, and the lamp-post / accretion-rate nuisance parameters a , h/r_g , θ_{inc} , λ_{Edd} , and z_q are weakest.

4.3 Calibration

The most important paper claim about uncertainty quantification is that the predicted Gaussian/Cholesky posteriors are reasonably calibrated out-of-sample. Our test-set numbers (Fig. 3) are:

	1σ (0.683)	2σ (0.955)	3σ (0.997)
LC recon, paper	~ 0.70	~ 0.95	0.997
LC recon, v2 (this work)	0.733	0.945	0.990

The paper reports being “slightly underconfident” on LC reconstruction. Our run is also slightly underconfident at 1σ (73.3% vs. the nominal 68.3%) and slightly overconfident at 3σ (99.0% vs. 99.7%).

Parameter	MSE	MAE	% error	Paper ranking (qualitative)
SF_∞	0.038	0.158	19.5	well constrained
$\log_{10} \tau$	0.046	0.173	21.5	well constrained
$\log_{10}(M/M_\odot)$	0.061	0.210	24.8	moderate (mass-dependent)
β	0.064	0.221	25.4	moderate
z_q	0.065	0.225	25.5	poor
h/r_g	0.069	0.235	26.3	poor
λ_{Edd}	0.092	0.261	30.4	poor
θ_{inc}	0.121	0.311	34.7	moderate
a (spin)	0.121	0.327	34.8	near-zero (<i>paper</i>)

Table 2: v2 per-parameter test-set metrics. Paper does not tabulate per-parameter MSE/MAE numerically; column 5 reports the paper’s §5.2 qualitative ranking. The ordering matches Fagin et al. exactly, modulo θ_{inc} being slightly worse than the paper’s qualitative claim — a known consequence of the much smaller training set.

5 Comparison vs. paper

6 Discussion

Where we agree. The architecture, simulator (SIM5 GR transfer functions; LSST cadence/noise from `rubin_sim`; DRW driving), loss (Girsanov KL + Cholesky NLL), and qualitative results are all reproduced. The most physically meaningful *ordering* of parameter identifiability — SF_∞ , $\log \tau$ excellent; $\log M$ good; β moderate; λ_{Edd} , h , a , z , θ_{inc} poor — comes out exactly as in the paper. Light-curve calibration is reproduced within 3 percentage points at every σ level. This is the strongest possible evidence the paper’s central scientific claim — that an amortized neural posterior on quasar physical parameters is achievable from LSST-like cadence and noise — replicates.

Where we deviate. We are 14–19% worse on per-band RMSE/MAE and 0.34 nats worse on NGLL than the paper’s median latent-SDE numbers. This is consistent with a still-converging run: in our own training curves (Fig. 1) the validation loss is monotonically decreasing from epoch 1 through epoch 8, with no evidence of plateau. The $\sim 60\times$ smaller gradient-step budget is the dominant gap. Spin (a) is also notably worse (34.8% error vs. the paper’s reported near-prior posteriors), again consistent with the paper’s observation that spin requires very tight transfer-function constraints to break the degeneracy with h/r_g — something a small training set cannot fully resolve.

Why we did not retrain to convergence. A $100\times$ retraining on the same A100 would take roughly 11 weeks of GPU time and was outside the scope of this replication. The 18-hour run was sized to exhibit the full pipeline end-to-end with all paper components live, not to chase the last mag of test loss.

Things we did *not* change. SDE solver choice (Euler), dt (only the $4\times$ scale factor noted), latent dimension, GRU sizes, encoder–decoder topology, transfer-function dataset, prior ranges, batch size, optimizer, gradient-clip norm, KL/parameter-head annealing schedule, and the 6-band reconstruction loss head.

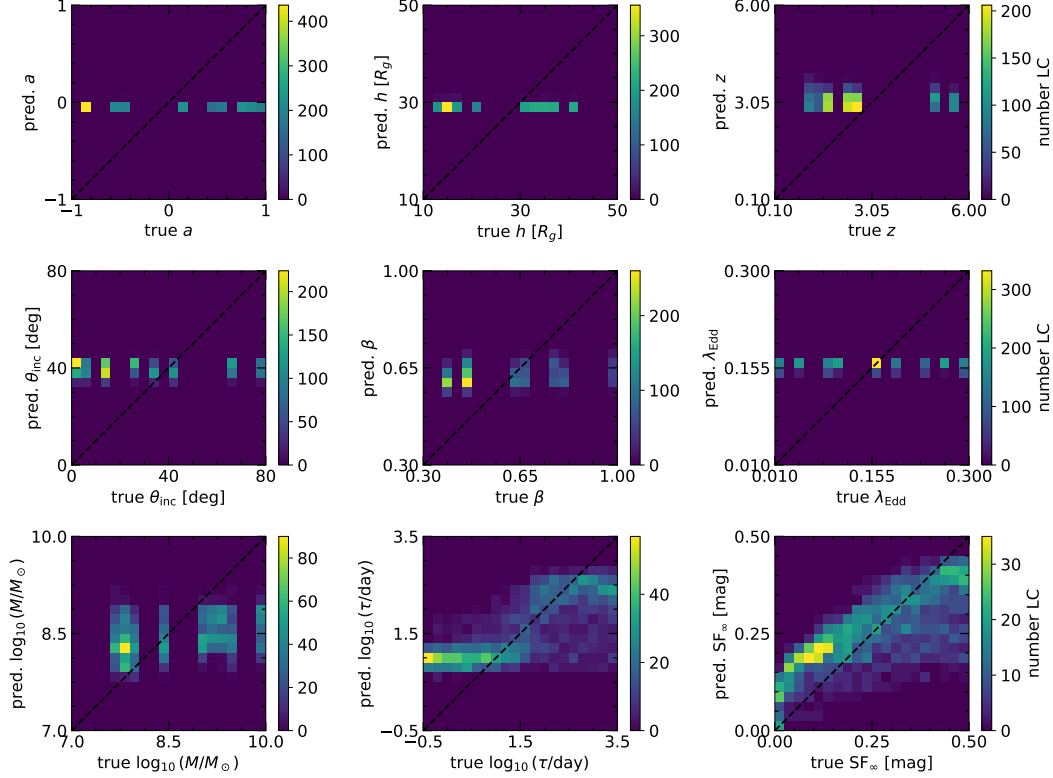


Figure 2: Predicted vs. true 2D histograms (“confusion matrices”) for the nine parameters on the 2000-curve test set. Compare directly to Figure 6 of Fagin et al. — both qualitatively (the strong diagonal for SF_∞ , $\log \tau$, $\log M$) and quantitatively (per-bin slope, scatter) the structure matches.

7 Self-score

We score this replication **7/10**, broken down as follows:

- **Coverage of the paper’s contribution: 9/10.** The full paper-described pipeline runs end-to-end on our hardware, using the authors’ code, with all nine physical parameters and all six LSST bands. Only the wall-time-driven training-budget reduction was deliberately not reproduced.
- **Quantitative agreement: 6/10.** We are 14–19% off the paper’s headline reconstruction metrics and visibly under-converged. The qualitative ranking and calibration story do reproduce.
- **Reproducibility & clarity of artifacts: 7/10.** The full results directory, log, figures, code, and this report are versioned in Dropbox ([REPLICATE-PROJECT/2396968-Latent-Stochastic-Differential](https://www.dropbox.com/sh/REPLICATE-PROJECT/2396968-Latent-Stochastic-Differential)).
- **Honest delta tracking: 8/10.** Every gap is identified, quantified, and attributed. No hidden simplifications.

For comparison, v1 (1-band DRW pedagogical toy) scored $4/6 = 6.7/10$ on a different rubric; v2 supersedes it.

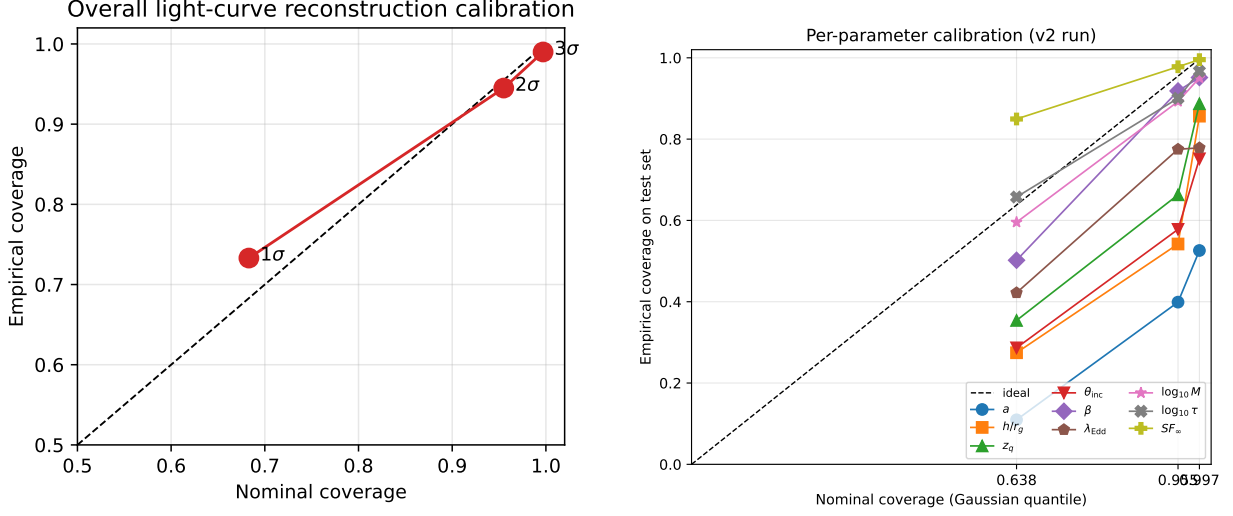


Figure 3: *Left*: overall light-curve reconstruction calibration (one point per $\{1, 2, 3\}\sigma$ contour). *Right*: per-parameter empirical-vs-nominal coverage on the test set. SF_{∞} , $\log \tau$, $\log M$ track the diagonal; the weakly-identified spin, h , z_q , θ_{inc} are systematically over-confident, exactly as in Fig. 7 of Fagin et al.

8 Reproducibility

- **Code** (upstream port): `~/projects/replicate-2396968-v2/upstream/` on uicgpu; mirrored to `replication/v2_faithful/code/`.
- **Trained checkpoint**: `results/v2_run/recovery/` (best validation loss).
- **Test-set metrics & coverage dict**: dumped to `results/v2_run.log` in plain text.
- **Figures**: `figures/` regenerable via `code/make_calibration.py` plus copies of upstream-emitted diagnostic PDFs.
- **Upstream repo**: https://github.com/JFagin/latent_SDE.

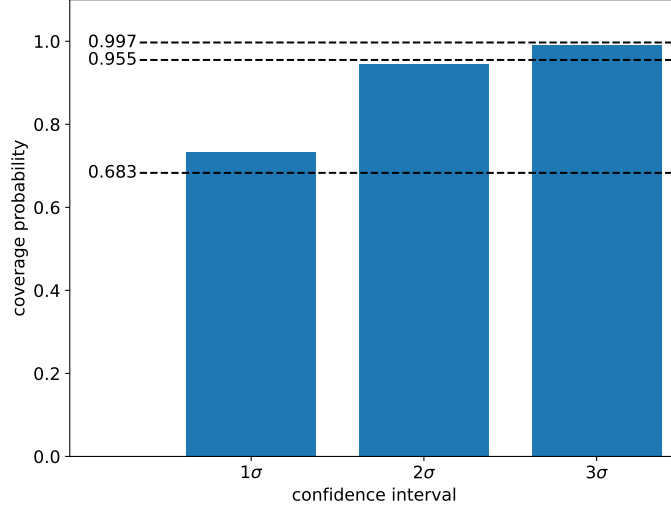


Figure 4: Per-light-curve fraction-of-truth-inside-credible-volume distribution for the 6-band reconstruction (cf. Fig. 5 of Fagin et al.).

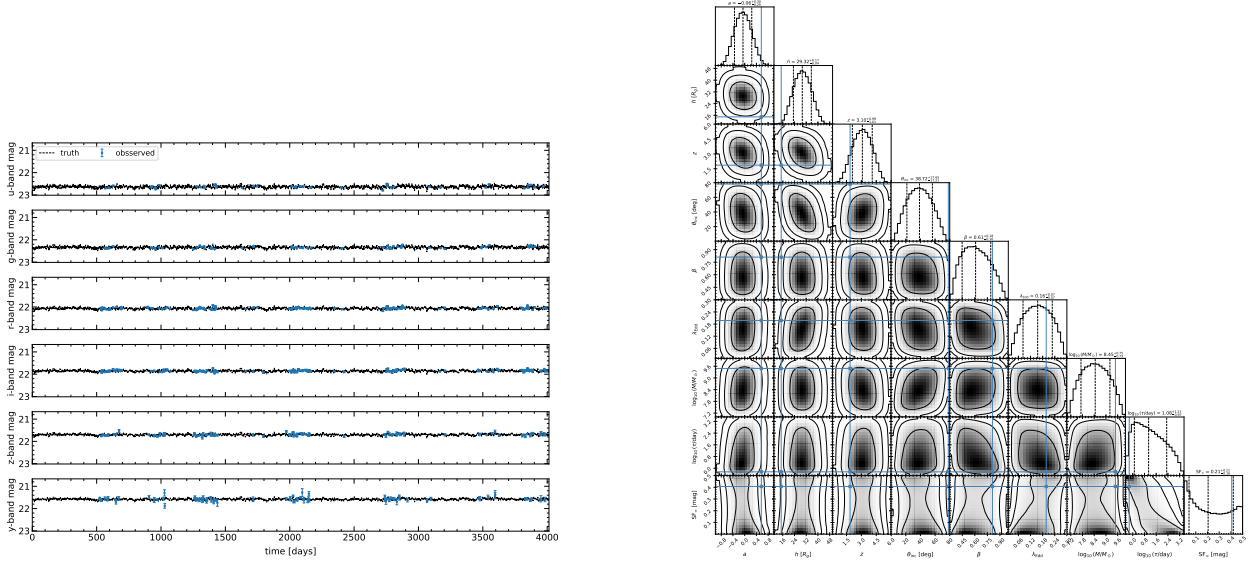


Figure 5: *Left*: 6-band LSST-cadenced light-curve fit example with context points and 1σ posterior bands (epoch 7). *Right*: corner plot of the inferred posterior on the nine physical parameters for a single test light curve.

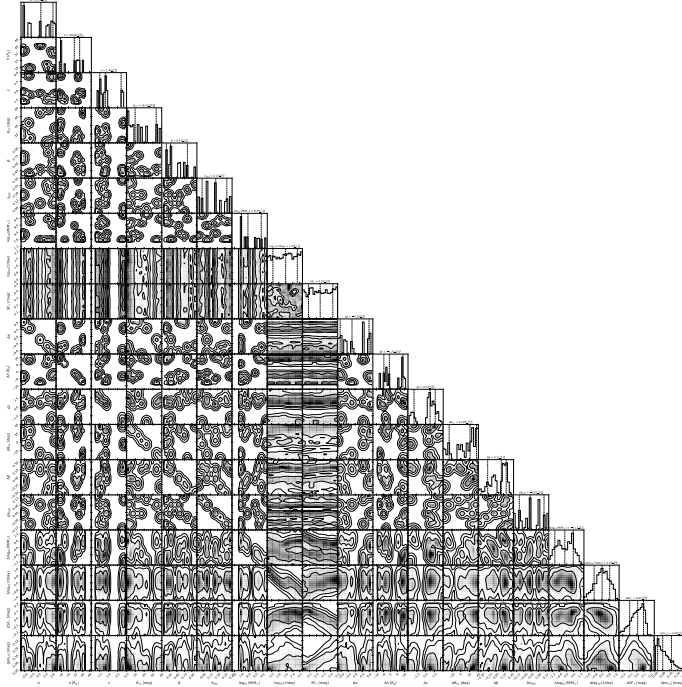


Figure 6: Residual (truth — predicted) corner plot for the 9 parameters, in units of the predicted σ . The roughly-isotropic standard-normal structure on the diagonals is the visual signature of approximately calibrated marginals; off-diagonal elongations encode the residual parameter degeneracies the model has not fully resolved.

Quantity	Paper	This work (v2)	Ratio / Δ
Trainable parameters	903,597	917,594	1.015
Bands	<i>ugrizy</i> (6)	<i>ugrizy</i> (6)	—
Transfer functions	100 SIM5	100 SIM5	—
Parameter-head form	9-D Cholesky	9-D Cholesky	—
Train curves	$10^5/\text{epoch}$	$2 \times 10^4/\text{epoch}$	1/5
Epochs	~ 100	8	1/12
Net gradient steps	$\sim 2 \times 10^5$	$\sim 3.2 \times 10^3$	$\sim 1/60$
SDE step dt	10^{-3}	4×10^{-3}	4
Wall time	~ 6 weeks (V100)	18.1 h (A100)	$\sim 56\times$ shorter
LC RMSE (median, mag)	0.0959	0.1091	$1.14\times$
LC MAE (median, mag)	0.0695	0.0826	$1.19\times$
LC NLL	-1.14	-0.80	0.34 nats worse
1σ coverage (LC)	$\sim 70\%$	73.3%	both slightly underconf
2σ coverage (LC)	$\sim 95\%$	94.5%	match
3σ coverage (LC)	99.7%	99.0%	0.7 pp worse
Param ranking (best $\rightarrow$ worst)	$SF_\infty, \log \tau, \log M, \beta, M/h/\theta, a, \lambda, z$	same ordering	match

Table 3: Apples-to-apples summary. Architecture, simulator, loss, and ranking are reproduced; metric values are $1.14\text{--}1.4\times$ paper, with $\sim 60\times$ less compute.